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Maritime Trade and Conflict: An Empirical Study of Nigerian Regions

Facoltà di Economia
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Alessandra Amicucci
ID number 1940795

Supervisor
Prof. Valerio Leone Sciabolazza

Co-Advisor
Prof. Michele Di Maio

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name (chairman, role)

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Author's email: email amicucci.1940795@studenti.uniroma1.it

Abstract

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Chapter 1

Introduction

The study how international economic integration affects stability in a country remains one of the central analysis in development economics. A large body of work looks at the effects of trade liberalization on growth and welfare in advanced economies, yet evidence for developing countries remains limited. In these settings, deep structural inequalities, weaker institutions, and a high reliance on primary commodities may shape very different transmission channels. In these contexts, trade represents an important channel through which external shocks influences local incomes and political stability. (UNCTAD, 2012). The existing empirical literature has mostly focused on commodity-price shocks as the principal external driver of conflict dynamics. In this branch of the literature, we can find studies like Dube and Vargas (2013) and Brückner and Ciccone (2010) showing that international price changes can influence the likelihood of civil conflict, although the direction of the effect depends on whether a country is an exporter or importer of the resource, as well as on the quality of local institutions. Parallel studies investigate how wars and instability disrupt trade and integration, thus the reverse relationship (Martin et al., 2008). Little evidence exists on how trade shocks themselves, rather than resource or conflict shocks, affect the probability of conflicts.
This analysiss attempts to be a first step in studying how exposure to trade influences subnational conflicts in Nigeria. Unlike previous studies that analyse price fluctuations or cross country variation in trade openness, the present analysis is based on data on port activity, maritime connectivity, and local socio-economic characteristics to measure how the diffusion of trade shocks across space affects the intensity of local conflict events. The aim is to move the focus from aggregate to micro variation, where the mechanisms can be observed more directly.

The effect of trade on conflict is theoretically mixed. Our research follows Berman and Couttenier (2015), they find that positive external income shocks can reduce conflict because it could improve economic opportunities and raise the opportunity cost of violence. At the same time, trade may increase inequality or shift incentives, as discussed by Amodio et al. (2024). This means that the net effect depends on local economic structure, the distribution of gains from trade, and the capacity of institutions to manage distributional tensions. In unequal economies, external shocks can therefore operate in two opposite directions: they

may stabilize societies by expanding livelihoods and smoothing consumption or destabilize them when benefits are accumulated by elites. We need spatial data to capure how trade exposure varies, and so, to identify which of these forces dominates.

Nigeria has different characteristic that made it a compelling candidate for examining these dynamics, as it is a coastal, rich in natural resources, and regionally fragmented country whose economy depends on external trade. Its geography combines well connected port areas with vast inland regions only loosely integrated into national and global markets. These heterogeneities make the foundation that we need for assessing how trade liberalization could influences conflict through income shocks. Moreover, Nigeria exhibits other characterists useful for the study such as marked disparities between East and West regions, ethnic fragmentation, and a recurrent pattern of localized violence, which together amplify the potential for uneven economic transmission. Coastal regions benefit from access to maritime infrastructure, while interior districts remain more isolated and vulnerable to economic exclusion. Understanding whether trade openness mitigates or exacerbates such disparities is crucial for interpreting the political economy of globalization in Africa. To capture these mechanisms empirically, we developed a spatial trade exposure index. It combines three components that can link global trade dynamics to local geography:

1. Trade shocks, measured as annual changes in port-level Container Port Throughput (CPT) from UNCTAD;
2. Connectivity of the port to other international one, proxied by the Liner Shipping Connectivity Index (LSCI) from UNCTAD, which is used to quantify each port's integration into global liner networks;
3. Proximity, expressed as the inverse of the minimum distance between each second level administrative unit (ADM2) and its nearest port.

In this way smaller distances and higher connectivity imply greater exposure to international trade shocks. This structure ensures that only areas geographically and economically close to ports experience significant variation in exposure when global trade conditions change. We interacted time varying national portual trade shocks ($CPT \times LSCI$) with fixed spatial heterogeneity (inverse distance), constructed this was the index respect the validation request by Goldsmith-Pinkham et al. (2020). It captures how fluctuations in global maritime activity translate into differential local effects across Nigeria's territory.

The contribution of this study is double. First, it provides a sub-national evaluation of trade shocks and conflict in Nigeria, integrating spatially detailed datasets on trade flows, port infrastructure, connectivity, conflict events, and environmental resources. This perspective allows the analysis to move beyond national averages and uncover within country heterogeneity in the economic and social consequences of trade. Second, the results reveal that the impact of trade liberalization is highly heterogeneous. Exposure to maritime trade tends to reduce conflict incidence less developed districts, but can reinforce instability in wealthier or rent-dependent areas, where new rents or inequalities emerge. This pattern highlights the central role of pre-existing economic and spatial inequalities in mediating the peace-promoting

effects of globalization. Beyond its empirical contribution, the thesis also offers a conceptual bridge between two strands of literature that have largely evolved in parallel: studies of external income shocks and conflict (Amodio et al., 2024; Berman & Couttenier, 2015) and analyses of spatial exposure to global markets inspired by gravity models (Anderson & van Wincoop, 2003). By linking these frameworks through a data-driven, geography-based measure of trade integration, the study provides new evidence on how international linkages permeate domestic political economies.

The remain of the thesis is organized as follows.Chapter 2 describes the data sources and the construction of key variables. Chapter 3 presents the conceptual framework and empirical strategy. Chapter 4 reports the main empirical results and chapter 5 concludes by summarizing the main findings.

Chapter 2

Data

In our empirical analysis, we combined data from different sources to derive a panel of subnational geographical (ADMIN2) units for the period from 2009 to 2019.

2.1 Depend variable

Conflict. Data on conflict event are drawn from the "Armed Conflict and Location Event" ACLED (2025) which cover conflict event from 1999 to 2024. The dataset provides the exact location of events, the disorder type, and other characteristics related to the conflict. (Raleigh et al., 2010)
In constructing the dataset, we focused on consumption-related, non-violent per se, conflict events rather than rapacity-based ones. Guided by both theoretical and empirical considerations given that standard trade models usually predict that trade openness influences a country's income in a positive way. However, globalisation has not always reduced inequalities in emerging countries, causing unequal distribution. (Dix-Carneiro & Kovak, 2023) As stated by Dube and Vargas (2013), an increase in income can reduce conflict, but on the other hand, higher income means there is more to fight over these resources. The existence of two opposing influences, opportunity cost and rapacity, suggests that some shocks may mitigate conflict, meanwhile other shocks could exacerbate it, depending on the relative strength of the two effects. The study of Amodio et al. (2024) shows that exposure to exports increasess simultaneously economic activity and political violence and McGuirk and Burke (2020) have shown that, if there is a positive price shocks, producers' revenues increase but consumers' real incomes decrease, enhancing the probability of protests or demonstrations. Conversely, Berman and Couttenier (2015) find that at the micro level positive income external shock, such us those generated by the trade, tend to have a pacifyng effect on conflict. In general, we can differenciate consumption-related mechanisms as more directly influenced by exogenous trade shocks, while violent conflicts over resources as affected by different dynamics, often linked to the strategies of armed groups. From an empirical point of view, limiting to consumption-related events also mitigates concerns about endogeneity: resource predation can in itself influence local trade flows, risking to reflect both cause and effect simultaneously.

The indipendent variable therefore measures the annual number of consumption-related conflict events for ADM2, that have been filtered to retain only: (i) subcategories of events not related to predation, such as protests, demonstrations, arrests, agreements or territorial changes not directly linked to resource predation. (ii) Events reported in narrative notes as disputes over consumer goods or livelihoods (keywords such as land, livestock, pasture, farm, grazing, oil, gold, looting). Events clearly associated with predatory motives (e.g., looting, armed clashes over resource extraction, kidnappings for ransom) are excluded.
After filtering, we limited the dataset to events occurring in coastal regions, then projected them into a consistent spatial reference system and overlaid them on Nigeria's second-level administrative units (ADM2). For each ADM2-year, the count of non-predatory conflict events is available, which serves as the outcome variable in my regression analysis.

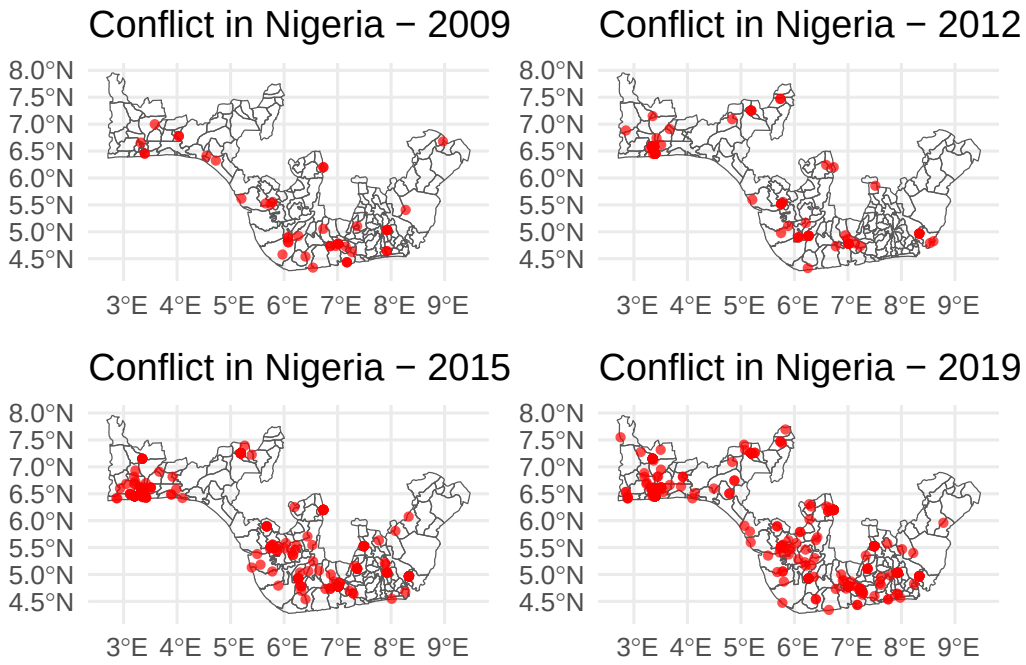


Figure 2.1 Nigeria conflict 2001- 2006- 2009- 2016

Figure 2.1 displays the spatial distribution of consumption-related conflict events across Nigerian coast in the years 2009, 2012, 2015 and 2019. The maps show a marked increase in conflicts between 2009 and 2019. In 2009, events were concentrated in a few clusters, while in 2019 they spread across most coastal regions, gradually extending westwards and inland. In 2015, there was a much more significant increase compared to 2012, and was greater than the gap between 2012 and 2009, suggesting an exponential growth dynamic.

Figure 2.2 shows the trend in the total number of conflict events per year (sum at ADM2 level). Focusing on the period 2009-2019, we can see that in the early

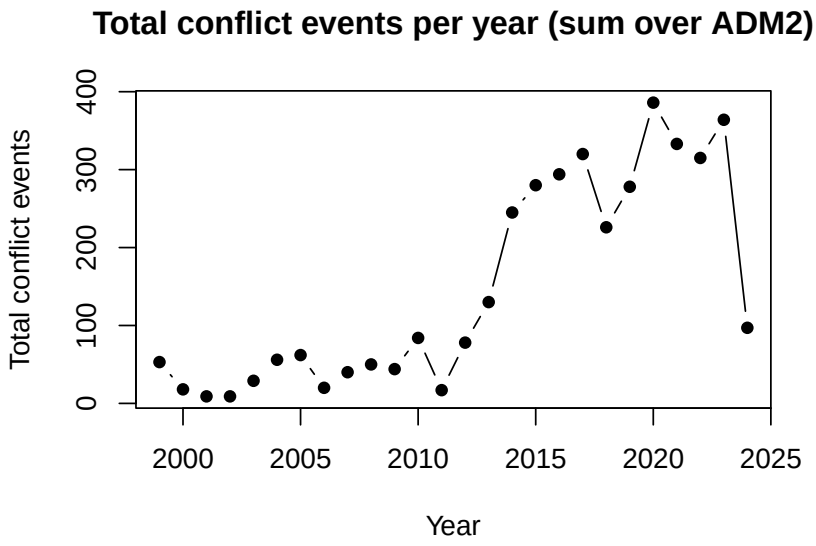


Figure 2.2 Conflict trend

years the number of incidents remained relatively low, but there has been a sharp increase since 2012, with progressive growth leading to a peak between 2015 and 2019. This trend confirms what has already been highlighted by the maps, namely a pattern of intensification and spread of conflicts over the decade.

2.2 Trade exposure

In addition to the outcome variable, the empirical design requires the construction of an index of trade exposure. This measure combines the temporal variation of a national trade shock with the spatial heterogeneity of local exposure, ensuring that both time and geography contribute to identifying variation.
The index was constructed based on the intuition that a trade exposure shock can significantly influence the dynamics of conflicts. In particular, we consider how the impact of trade shocks varies geographically depending on the distance from trade ports: as highlighted by Berman and Couttenier (2015), regions closer to ports are in fact more exposed to external shocks. Following this intuition, we constructed an index that combines the measure of national connectivity with a local proxy for geographical accessibility to ports.

$$index = Shock \times LSCI \times 1/(distminnorm + \varepsilon) \tag{2.1}$$

Shock. Here, the shock is represented by Container Port Throughput, a dataset compiled by UNCTAD that estimates the total number of containers handled, by country, expressed in twenty-foot equivalent units; LSCI stands for Liner Shipping Connectivity, which measures a country's integration into global liner shipping networks, also from UNCTAD; finally, there is the subnational variable, which is the

inverse of the normalized minimum distance to the nearest commercial port and ε is a very small positive constant introduced to avoid division by zero.

The main idea is that being close to commercial ports makes it easier to get to international markets. In other words, areas near ports are more exposed to international trade, while areas far away are relatively isolated. This insight is consistent with empirical evidence that regions closest to ports are most affected by global shocks, whether positive or negative, that are transmitted through trade (Berman & Couttenier, 2015; McGuirk & Burke, 2020). Similar to the quasi-experimental approach by Amodio et al. (2021), who exploited Israel's 2008 ban on dual-use imports to the West Bank as an exogenous shock to trade, the present index captures how sudden or structural changes in trade integration may affect local economic and conflict dynamics. While their evidence shows that trade restrictions depress wages and increase violence, here the focus is to assess whether greater maritime accessibility and trade intensity can mitigate conflict in a developing-country context such as Nigeria.

Linear Shipping Connectivity. The national component of the index is represented by the Liner Shipping Connectivity Index (LSCI), developed by UNCTAD (2025a) and frequently used in maritime trade literature as shown in Mishra et al. (2021) and Fugazza and Hoffmann (2017). The LSCI measures a country's structural capacity to integrate into global trade, based on variables such as the number of liner services, active operators, ships employed, ship size and total capacity in Twenty Foot Equivalent Units (TEU). It is internationally recognised as an indicator of trade accessibility and competitiveness, particularly relevant in developing countries where maritime transport is the main route for international trade. Empirical studies confirm that greater maritime connectivity reduces trade costs and promotes growth: for example, Fugazza and Hoffmann (2017) estimate that the absence of direct connections can reduce the value of bilateral exports, while the Review of Maritime Transport UNCTAD (2012) shows that improvements in the LSCI ranking are systematically associated with higher trade volumes and greater market access.

Foreign trade shock is measured using Container Port Throughput (CPT), a dataset compiled by UNCTAD (2025b) that reports the annual number of containers handled in each country's ports, expressed in TEU. CPT is used as an indicator of the intensity of maritime trade flows, as most international trade, in terms of volume, takes place by sea (Ayesu & Boateng, 2024). Meanwhile the LSCI captures the potential capacity of a national maritime transport network, the CPT reflects actual containerised traffic and thus provides a more direct measure of exposure to global supply and demand shocks. Empirical studies confirm that changes in CPT have tangible macroeconomic effects. For example, Ayesu and Boateng (2024) demonstrate that increased containerised traffic significantly boosts employment in African economies with maritime ports, highlighting its role as an engine of economic development. At the same time, the OECD/ITF report by Ducruet et al. (2014) points out that throughput also reveals structural vulnerabilities: in ports where efficiency is low, high levels of throughput can amplify the impact of external shocks due to congestion and longer periods of stay. From a policy perspective, UNCTAD

UNCTAD (2012) recognises CPT as a key indicator of port development performance in emerging economies, highlighting its value as a standard measure of exposure to international trade. In this analysis, we consider the CPT as an exogenous source of variation in maritime trade flows which, interacting with measures of national connectivity (LSCI) and local port accessibility (distance from port), captures the heterogeneous transmission of global trade shocks to subnational regions.

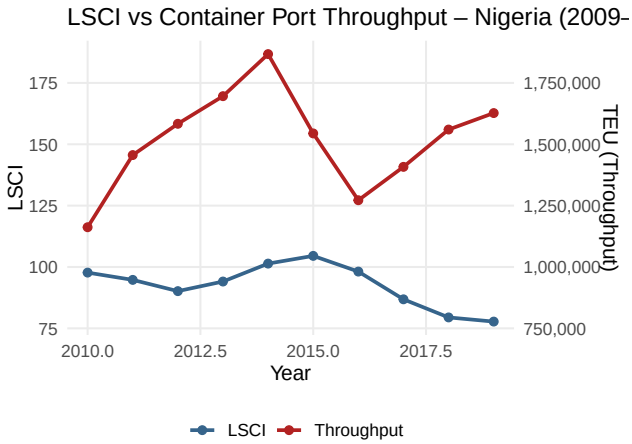
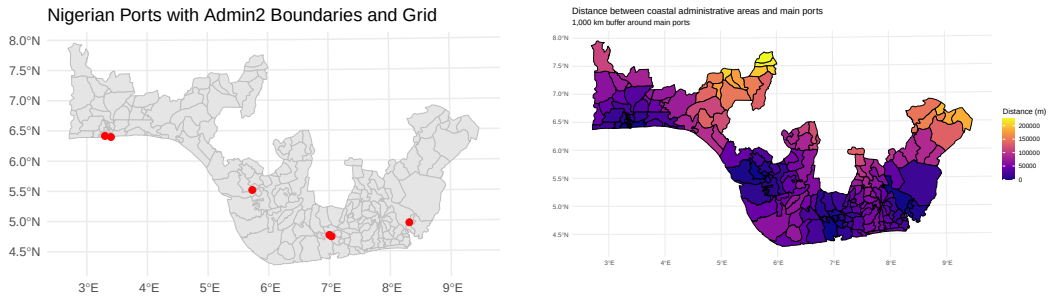


Figure 2.3 LSCI and CPT trend

Figure 2.3 presents the trend of the LSCI and CPT in Nigeria over the period 2009–2019. The two indicators show a similar trend: sustained growth until 2014–2015, followed by a contraction phase and a partial recovery in recent years. Their co-movement confirms that LSCI and CPT capture complementary dimensions of the same trade dynamic: the former reflects the structural capacity for integration into global trade, the latter the actual measure of the volume of containers handled. For this reason, both are included in the construction of the trade shock index used in the empirical analysis.

Distance to trade ports. The role of distance from ports is crucial to understanding the heterogeneity of trade shocks. Following the intuition of gravity models Anderson and van Wincoop (2003), geographical distance remains a central determinant of trade costs and access to markets. In our framework, minimum distance from ports is therefore employed as a measure of local exposure to international trade shocks. (Anderson & van Wincoop, 2003) According to Berman and Couttenier (2015), regions located closer to major seaports are more directly integrated into international trade networks and therefore more strongly affected by external income shocks. Their findings show that positive shocks reduce the likelihood of conflict within these open regions, while the effect largely disappears in remote inland areas. This pattern suggests that better access to trade increases local incomes and raises the cost of using violence, thereby mitigating conflict. Following this idea, we incorporated distance from ports into the construction of my index. The logic is that Nigerian districts closer to seaports are more exposed to changes in global trade flows and should therefore exhibit a stronger response to consumption-related

conflicts in the event of external shocks than districts located further inland. For each coastal ADM2, the distance from the nearest port was calculated, converted into kilometres and then normalised through a min–max transformation. To ensure comparability across units and to give greater weight to areas close to ports, the normalised distance was inverted so that higher values indicate stronger exposure to trade. Figure 2.4a shows the position of the six main ports overlapped for the ADM2 subnational regions. These ports correspond to those officially listed by the Nigerian Ports Authority and identified in the IMF PortWatch database as handling the largest shares of national maritime trade activity. Starting from those ports the normalised minimum distance between each coastal administrative unit and the nearest commercial port was calculated. Figure 2.4b report the spatial distribution within a 1,000 km buffer. We can see that the coastal areas immediately adjacent to the ports, in dark purple, are naturally much more accessible, while the distance increases rapidly as one moves inland, highlighted by the lighter shades. This geographical heterogeneity forms the basis for the variable, which assigns greater exposure to areas closer to ports and progressively reduces that of more remote regions.



(a) Main trade ports **(b)** Buffer around main ports

Figure 2.4 Comparative maps: location of ports(a) and minimum distance from them(b)

While the LSCI index measures Nigeria's overall capacity to engage in global maritime trade, the minimum distance to ports is a reflection of regional disparities in the actual access to these connections. By integrating these two dimensions with the external shock represented by CPT, the index reflects both the macro structure of the port system and local differences in exposure. In this way, not only are general changes in the country's position within maritime networks captured, but also how such shocks are absorbed differently depending on geographical proximity to ports. The combination of these elements allows us to move from a uniform national perspective to a more detailed one, highlighting the territorial heterogeneity of the effects of maritime trade and their relevance to the dynamics of conflicts.

2.3 Control variable

Nighlights Night-time lighting (NTL) data are widely used in the literature as an indicator of economic activity, especially in contexts where conventional economic statistics are scarce or unreliable. The underlying idea is that artificial night-time lighting is strongly correlated with income levels, infrastructure and access to energy, thus serving as a reliable indicator of local development (Amodio et al., 2024; Chen & Nordhaus, 2011; Henderson et al., 2012). In the case of Nigeria, night-time lights have been used to assess access to electricity and regional inequalities, confirming their strong association with patterns of socio-economic development. To construct this variable, we relied on the Harmonised Global Night-time Light Dataset (1992-2018) developed by Li et al. (2020), which combines the DMSP-OLS series with the latest VIIRS observations into a consistent time series. The harmonisation procedure corrects for satellite drift, blooming, and sensor inconsistencies, thus ensuring temporal comparability. The dataset is available in yearly composites and can be aggregated spatially at multiple scales, making it particularly suitable for integration with ADM2-level units.For each coastal ADM2 and for each year, we calculated the logarithm of night-time light intensity to reduce asymmetry and facilitate interpretation in regressions. In addition, we defined a dummy variable equal to one for ADMs with values above the 75th percentile of the log-NTL distribution within the year. This distinction allows us to test whether trade shocks have differential effects between more and less economically developed regions.

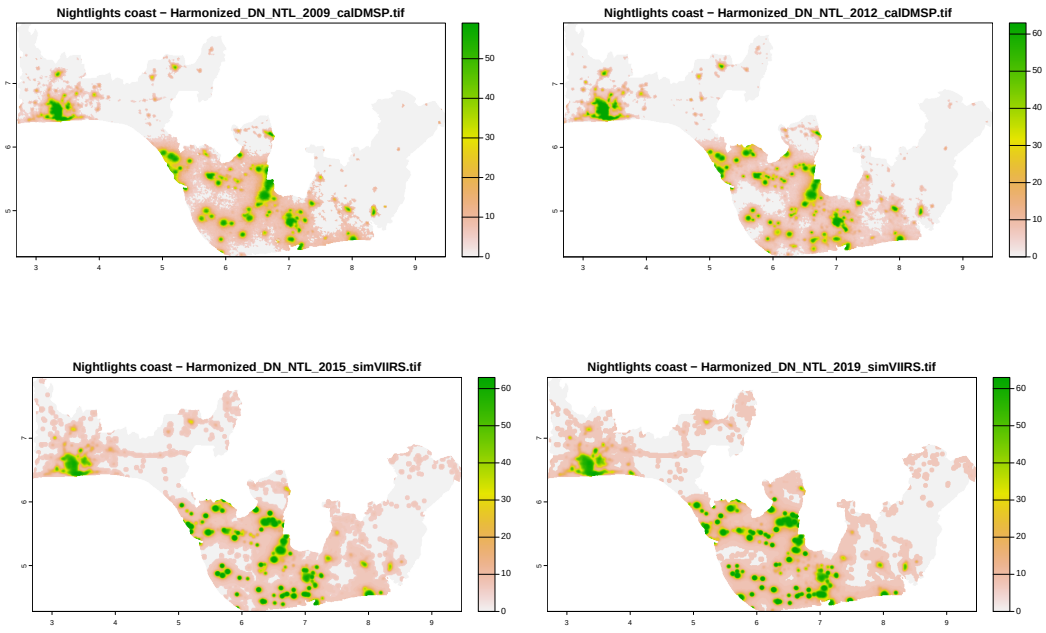


Figure 2.5 nightlights

Figure 2.5 shows the spatial distribution of night-time light intensity in Nigeria's coastal ADM2s. Light concentration is clearly clustered around Lagos, Port Harcourt,

and other industrial centres, while rural areas show very limited brightness, consistent with the literature on energy poverty in Nigeria.

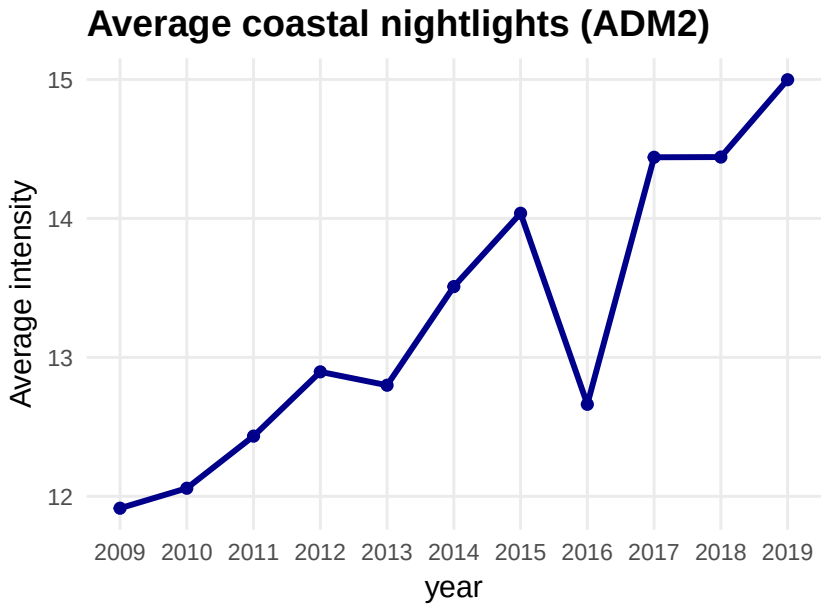


Figure 2.6 Average coastal nightlights

Moving on to temporal dynamics, Figure 2.6 tracks the average intensity of night-time lights over the period 2000–2018. The trend reveals an overall increase in luminosity, reflecting the gradual urbanisation of Nigeria, but also a significant decline around 2013 and 2016. These fluctuations are consistent with the technical issues documented in the harmonised dataset. In particular, Li et al. (2020) notes that the decline observed in the period 2015–2016 is also present in independent VIIRS composites and attributes it to sensor fluctuations and the transition from DMSP to VIIRS rather than an actual economic contraction. Similarly, Islam (2024) highlights the challenges posed by blooming, overglow, and inter-sensor calibration in the Nigerian context, emphasising that apparent discontinuities in time series may reflect data artifacts rather than real changes in development. It is important to note that such noise has been highlighted in previous studies as a limitation of remote sensing for socio-economic monitoring, but the harmonisation procedure mitigates much of the distortion. We analyse the fluctuations in Appendix A
In summary, night-time lights provide a proxy for local economic activity. The expectation is that areas with lower night-time light intensity (i.e. poorer and less developed regions) should be more sensitive to foreign trade shocks, as consumption disruptions are more likely where households face tighter constraints on basic goods and services.

2.4 Heterogeneity

Geographical and sector characteristics. To account for spatial and structural differences across Nigeria's coastal economy, we classified ADM2 units and ports along two complementary dimensions. First, East–West regions were defined according to the median longitude of ADM2 centroids, distinguishing the eastern coastal belt—centered around Port Harcourt and Calabar—from the western belt dominated by Lagos and its surroundings. This partition captures historical and economic disparities in port activity and industrial development. Second, sectoral and trade-flow specializations were derived from the IMF PortWatch database, which provides information on the predominant categories of goods traded through each port. Following the data availability since 2019, ports were classified as either vegetable-oriented or chemical-oriented, the two main product groups after minerals, and as predominantly import- or export-oriented depending on the share of incoming and outgoing containerized flows. These distinctions are used later in the heterogeneity analysis to test whether the pacifying effect of trade exposure differs by regional position, sectoral composition, or trade direction. We report the sources for all these variables in Appendix B

Water availability We expect that regions with greater water availability are those where economic activity is higher and more connected to national markets. Since the vegetable sector represents one of the main traded categories in Nigerian ports, second or third depending on the port, water resources are an essential input and thus a relevant proxy for agricultural income. For this reason, we use water availability as an indicator of agricultural activity and local economic potential (FAO & IIASA, 2020). The data come from FAO–GAEZ, which provides raster information on the mean quantity of water resources (or hydrological balance) available in each geographical unit. We calculate the mean value for every ADM2 area, obtaining a spatially continuous and relatively stable characteristic of the territory. This variable is then used to test how the pacifying effect of trade varies with the underlying resource environment.

Chapter 3

Methodoloy

We expect the economic impact of trade shocks to vary across different ADM2 units, with areas most exposed to international trade experiencing fewer consumption-related conflict events. Following this intuition, the index presented in the chapter Data was constructed, combining two national components that vary over time with a fixed spatial component. This approach is consistent with consolidated practices in the literature that combine aggregate shocks with sub-national heterogeneity (Autor et al., 2013).

The mechanism linking trade exposure and conflict is consistent with the opportunity-cost and grievance framework discussed by Amodio et al. (2021). Their evidence suggests that economic integration and access to trade can shape local conflict dynamics by altering individuals' economic incentives and perceptions of welfare. In this thesis, the focus is on the opposite mechanism: instead of trade restrictions, we study whether higher trade openness and proximity to maritime ports, which expand local economic opportunities, can mitigate the incidence of violence. The same channels highlighted by Amodio et al. (2021) are therefore expected to operate in reverse, with improved trade connectivity reducing both grievance and the opportunity to engage in conflict.

3.1 Conceptual framework

To better capture the mechanisms through which trade exposure affects social stability, we construct a specific subset of conflict events reflecting consumption- or income-related tensions rather than resource-seeking or political violence. Starting from the ACLED dataset (ACLED, 2025), we select events classified as protests or demostrations, the types less likely to have a link with resource predation. In particular, we identify income-related events through a keyword-based text filter, selecting those that mention terms associated with basic-needs or livelihood shocks, such as food, fuel, price, subsidy, market, ration, maize, rice, bread, or tax. Conflict incidence can be represented as (3.1) where Y_{it} captures opportunity-income effects (the effect of trade in income, reducing conflict) and R_{it} captures rent-seeking incentives (the opposite effect of trade that incentivises conflict).

$$Conflict_{it} = f(Y_{it}, R_{it}) \tag{3.1}$$

Our index captures exogenous variation in the income effect transmitted through trade shocks, holding the incentives largely constant by excluding rapacity-related events.

3.2 Identification Strategy

Identification relies on within district (ADM2) variation in exposure to national trade shocks over time. Because the trade shock (CPT × LSCI) is common across units while the distance is time-invariant, only the interaction between these components generates differential effects once ADM2 and year fixed effects are included. In our case, the CPT was delayed by one year to mitigate possible simultaneity issues, while the LSCI represents the degree of integration of the Nigerian port system into global maritime networks. Spatial heterogeneity is introduced through the minimum distance from ports, which serves as a proxy for accessibility to international markets. Following the intuition of (Berman & Couttenier, 2015), regions located closer to ports are more directly integrated into international trade networks and therefore more strongly affected by external shocks. In our context, this motivates the use of minimum distance from ports as a measure of local exposure, assuming that trade shocks are transmitted more intensely to areas with better access to maritime infrastructure. In this sense, the empirical design is consistent with the validity conditions discussed by (Goldsmith-Pinkham et al., 2020)
The empirical model also includes both spatial (ADM2) and temporal (year) fixed effects. In this way, given the same national shock, only differences in proximity to ports generate variation between units, allowing the heterogeneous effect of trade shocks on conflict to be isolated. To account for spatial heterogeneity the distances from ports where normalised. Since the relationship between trade exposure and conflict is expected to decrease with distance, the raw distances (measured in kilometers) were first inverted to capture proximity rather than remoteness. To ensure comparability across regions, the proximity measure was then rescaled between 0 and 1, so that values closer to 1 represent areas with higher accessibility to maritime trade. This transformation preserves relative spatial differences while avoiding scale distortions that could bias the exposure index.

3.3 Empirical Strategy

All models are estimated using OLS with two-way fixed effects (ADM2 and year). Standard errors are clustered at the ADM2 level to account for serial correlation within units over time. The empirical strategy consists of three steps. *First*, we estimate the basic model in two specifications: an aggregate version, in which the exposure index is considered in its overall form (Equation(3.2)), and a "split" version, which distinguishes administrative units based on the intensity of economic development measured by night-time lights (Equation (3.3)). This allows us to analyse both the average effect of the trade shock and the differences between more and less developed areas.

$$Conflict_{it} = \beta Index_{it} + \gamma log(Nightlights_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \tag{3.2}$$

$$Conflict_{it} = \beta_1 Index_{it}^{HigNL} + \beta_2 Index_{it}^{LowNL} + \gamma log(Nightlights_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \tag{3.3}$$

Second, we perform a falsification test to verify that the results do not simply reflect a spurious correlation between conflict and geographical proximity to port infrastructure. To this end, we reconstruct the index to include all ports in Nigeria, not just commercial ones. The idea is that if the effect were linked exclusively to physical proximity to a port, it should also emerge with this broader measure. Conversely, if the association is really driven by trade shocks, the results should only appear when considering ports that are actually relevant to international trade.
Third, beyond the baseline model, we explore whether the trade–conflict relationship varies across key structural dimensions. Following the idea that the effects of economic integration depend on pre-existing territorial and social characteristics ((Berman & Couttenier, 2015)), we analyse the heterogeneity of the effects across region, sector and environment. In particular, we distinguish between (i) eastern and western regions, (ii) the predominant type of industry traded, (iii) the predominant trade flow and (iv) water availability. These specifications allow us to investigate the differentiated transmission of trade shocks depending on the structural conditions of the territories. Equation 3.4 extends the baseline model to account for heterogeneity across structural or environmental dimensions. The interaction term captures whether the effect of trade exposure differs depending on regional location or sectoral specialization. A negative and statistically significant coefficient on β_2 would indicate that the pacifying effect of trade is stronger in the corresponding subgroup.

$$Conflict_{it} = \beta_1 Index_{it} + \beta_2 (Index_{it} \times H_i) + \gamma \log(Nightlights_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \tag{3.4}$$

Equation 3.4 account for water availability heterogeneity. $HighWater$ and $LowWater$ are dummies equal to one for ADM2 units above or below the median of the FAO–GAEZ water-availability distribution. In this analysis the coefficients $\beta 2$ and $\beta 3$ capture how the trade–conflict relationship differs across regions with different quantity of water resources

$$\begin{aligned} Conflict_{it} = {} & \beta_1 Index_{it} + \beta_2 (Index_{it} \times HighWater_i) + \beta_3 (Index_{it} \times LowWater_i) \\ & + \gamma \log(Nightlights_{it}) + \mu_i + \lambda_t + \varepsilon_{it} \end{aligned} \tag{3.5}$$

Chapter 4

Results

In what follows, we present the results. Table 4.1 reports the estimates of the main coefficient from Equation (3.2) and (3.3).

Table 4.1 Correlation between port shocks and conflict: commercial vs all ports

	(1) Commercial ports	(2) Commercial ports	(3) All ports	(4) All ports
indice2_main	-1.01e-07		-6.77e-08	
	(6.30e-08)		(4.29e-08)	
indice2_highNL		-1.12e-07		-9.32e-08
		(7.80e-08)		(6.02e-08)
indice2_lowNL		-5.61e-08***		-1.81e-08
		(5.43e-09)		(1.82e-08)
log(nightlight)	-0.879***	-0.878***	-0.870***	-0.867***
	(0.249)	(0.248)	(0.249)	(0.249)
Observations	1,611	1,611	1,611	1,611
Adj. R^2	0.662	0.662	0.661	0.662
Within R^2	0.028	0.029	0.024	0.027
RMSE	1.806	1.805	1.810	1.807

Fixed effects: ADM2 and year. SE clustered at ADM2 level.

358 observations dropped due to RHS NA. Signif.: *** 0.01, ** 0.05, * 0.1.

It shows the results of the baseline specification, comparing the commercial ports with the broader set of Nigerian ports. Column (1) and (2) focus exclusively on the commercial ports, while Columns (3)–(4) include all ports as a falsification test.
Starting from Column (1), results indicates that the aggregate coefficient on the exposure index is small and not not statistically significant. However, when the sample is split to study the heterogeneity between high- and low-nightlight areas, as in Column (2), a strong heterogeneity emerges. his split distinguishes ADM2 units by local economic development, proxied by night time lights (as detailed in the Data section): where the areas with higher income are the ones with values above the 75 percentile (Amodio et al., 2024; Henderson et al., 2012).
In the areas with reatevely low levels of nightlights, the effect of the commercial shock is negative and statistically significant, suggesting that a greater exposure maritme trade reduces the likelihood of consumption-based conflicts In contrast the coefficient is close to zero in the more developed area. This model is coherent

with the intuition that the external trade shocks have stronger effect where the local economies are less diversified and more vulnerable to changes in global market integration (Berman & Couttenier, 2015). Thus distance from commercial ports is particularly important in the less developed regions, amplifying the regulatory effect of the trade shocks. This finding is especially relevant given that less developed areas are also more likely to have conflict (Brückner & Ciccone, 2010; Collier & Hoeffler, 2004)

Columns (3)–(4) serve as a falsification test. If the relationship was determined solely by proximity with any port, not only the commercial one, we would expect similar effects when all ten Nigerian ports are included. Columns 3 and 4 extend the analysis, considering all nigerian ports, regardless their commercial activity. The results show weaker and not statistically significant coefficients, both for the aggregate and the splitted samples. The absence of an effect reinforces the interpretation that the mechanism is specificaly linked to international trade flows, rather than mere geographic proximity.

Overall, the results suggest that the link between trade and conflict is neither mechanic nor omnipresent, it rather depends commercial relevance of ports and the level of local economic development. The coefficients support the idea that international integration can provide stabilising benefit in the peripherical regions, and so the less developed ones economically speaking,while its marginal impact in more developed areas remains limited.

4.1 Heterogeneity

After establishing the baseline relationship, we explore whether the estimated effects vary across Nigeria's main macro-regions and according to local social fragmentation. Table 4.2 combines these dimensions of heterogeneity: Panel A shows how the trade–conflict link differs across regions and sectors, whereas Panel B reports the corresponding nightlight averages, illustrating the underlying economic disparities between groups.

Geography. Columns (1) investigates geographic heterogeneity: the index is interacted with East/West dummies defined by the median longitude of coastal ADM2 centroids (details in Appendix B.1). The estimated effect is more negative and statistically stronger in the eastern belt—areas that are poorer and more connected to maritime activity—consistent with a vulnerability-to-trade mechanism. A detailed discussion of the spatial correlation between the macro-regional partition, together with corresponding maps, is reported in Appendix B.1.

These patterns mirror the spatial distribution of economic development (Panel B). Nightlight data shows that western ADM2s exhibit higher and more dispersed nightlight values—largely driven by the Lagos metropolitan area—while eastern units are uniformly dimmer, reflecting a more rural and less industrialized eco-

Table 4.2 Regional and sectoral heterogeneity (combined)

Panel A. Index effects

	East vs West	Chemicals vs Vegetables	Export vs Import	Tribal	
indice2_est	-1.63e-07 *				
	(8.24e-08)				
indice2_west	-7.45e-09				
	(4.76e-08)				
indice2_main_veg		-1.63e-07 *			
		(8.24e-08)			
indice2_main_chem		-1.56e-07			
		(1.27e-07)			
indice2_main_exp			2.13e-09		
			(3.29e-08)		
indice2_main_imp			-2.55e-07 ***		
			(5.84e-08)		
indice2_lowNL				5.58e-05	-5.6e-08***
				(4.42e-05)	(5.47e-9)
indice2_highNL				-1.12e-07	-1.119e-07
				(7.81e-08)	(7.81e-08)
indice2_lowNL_tribe				5-59 e-05	
				(4.16e-05)	
indice2_lowNL_frac					3.92e-05
					2.90e-05
Controls					
log(nightlight)	-0.884 ***	-0.884 ***	-0.884 ***	-0.877***	-0.877***
	(0.2493)	(0.2493)	(0.2470)	(0.248)	(0.248)
Observations 1,611	1,611	1,611	1,611	1,611	1,611
Adj. R^2	0.665	0.665	0.670	0.662	0.662
Within R^2	0.0361	0.0361	0.0499	0.0288	0.0288
RMSE	1.799	1.799	1.786	1.805	1.805

Panel B. Average nightlights (mean / median / sd)

Group	Mean NL	Median NL	SD NL	N
East	9.78	6.57	10.4	979
West	16.8	8.91	18.9	990
Chemicals	16.5	8.41	19.2	957
Vegetables	10.3	6.76	10.6	1,012
Export	16.4	9.17	17.9	1,199
Import	8.48	5.83	9.52	770

Fixed effects: ADM2 and year. SE clustered at ADM2 level. 358 obs. dropped due to RHS NA. Signif.: *** 0.01, ** 0.05, * 0.1. NL = nightlights (mean annual radiance, harmonized DMSP-OLS/VIIRS). ADM2-level averages, 2009–2019.

nomic structure. In this case, the gap between richer and poorer areas (as proxied by nightlights) is particularly pronounced. Detailed maps, descriptive statistics of night-lights by region, and additional robustness checks are reported in Appendix B.2.

Sector. Columns (2)-(3) reports the sector heterogeneity: the index is interacted with two different sets of dummies: (i) the predominant type of industry traded through each port — either chemical or vegetable products (the two main sectors after minerals, based on data from IMF PortWatch, available since 2019 and used here as a proxy); and (ii) the predominant trade flow orientation, that is, whether the port handles mainly imports or exports (also based on data from IMF Portwatch). The estimated effect is negative and statistically significant for

vegetables and imports. For presentation purposes, we refer to the coefficients as index_*veg* and index_*chem* (or index_*exp* and index_*imp*), where the former represents the baseline effect and the latter includes the interaction term. The stronger and statistically significant effects observed for vegetable and import-oriented ports suggest that not only proximity and structural economic conditions matter, but also the nature and direction of trade flows. The results are consistent with previous findings that trade in basic consumption goods, such as agricultural products, can mitigate conflict by stabilizing local prices and incomes ((McGuirk & Burke, 2020)). The effect for vegetable-oriented ports is also consistent with what is shown by Amodio et al. (2024): regions with lower nightlight intensity, typically linked to agricultural activity, are more vulnerable to economic shocks, suggesting that the welfare channel of trade is particularly relevant in such contexts. Similarly, import flows—particularly of essential goods—may enhance welfare and reduce consumption-related tensions (Brückner & Ciccone, 2010; Dube & Vargas, 2013). Overall, these patterns reinforce the idea that trade integration contributes to stability primarily through welfare-improving, rather than resource-based, channels.
Results are aligned with the underlying economic structure summarized in Panel B, where vegetable and import-oriented ports correspond to areas exhibiting more dispersed nightlight distributions, a pattern typically associated with more rural and less industrialized economies. Although the import/export distinction also reflects structural differences in local economies, import-oriented ports show a systematically stronger effect than all other categories. Overall, these patterns reinforce the idea that trade integration contributes to stability primarily through welfare-improving, rather than resource-based, channels.

We next investigate whether the pacifying effect of trade exposure varies with local resource endowments. Water availability, derived from the FAO–GAEZ database (FAO & IIASA, 2020), serves as a proxy for agricultural potential and economic activity, given its close link to crop productivity and the prominence of vegetables trade in Nigerian ports.

The intuition is that in ADM2 units with lower water availability, typically more fragile and rural, the trade index should reduce conflict by stabilizing consumption and providing alternative income channels. This mechanism is consistent with evidence that economic integration can mitigate consumption-related conflicts by improving access to markets and smoothing shocks (Amodio et al., 2024).

However, the empirical results in Table 4.3 reveal a more complex pattern. Contrary to expectations, the effect of trade exposure turns positive in areas with higher water availability—where agricultural and economic activities are already developed. This suggests that, rather than amplifying stability, trade may in these contexts reinforce existing inequalities or intensify competition over profitable markets (Berman et al., 2017; Michalopoulos & Papaioannou, 2013). Overall, the findings indicate that trade openness exerts a stabilizing influence mainly in less developed and resource-constrained local economies, a result in line with broader evidence on the heterogeneous development impacts of market integration (Amodio et al., 2024; Dix-Carneiro & Kovak, 2023).

Table 4.3 Water Availability: Levels & Heterogeneity

	(1) year FE	(2) year FE	(3) ADM2+year FE	(4) ADM2+year FE
Index (overall / highNL / lowNL)	-3.59e-08	-3.95e-08 (highNL)	-1.327e-07*	-6.40e-08
	(1.21e-07)	(1.30e-07)	(6.09e-08)	(8.76e-08)
	–	-2.16e-08 (lowNL)	–	–
		(1.04e-07)		
Index × High water (median)	–	–	1.076e-07	–
			(6.88e-08)	
Index × Low water (median)	–	–	-1.40e-08	–
			(3.17e-08)	
Index × Water tercile: Medium	–	–	–	-2.597e-07**
				(8.75e-08)
Index × Water tercile: High	–	–	–	3.77e-08
				(9.38e-08)
Log Nightlights	0.902599**	0.902743**	-0.864138***	-0.868460***
	(0.199140)	(0.199352)	(0.245164)	(0.246453)
Water availability	-0.037963**	-0.037975**	–	–
	(0.010469)	(0.010490)		
Observations	1620	1620	1620	1620
Adj. R^2	0.102	0.101	0.664	0.668
Within R^2	0.0816	0.0816	0.0331	0.0454
RMSE	3.116	3.116	1.797	1.785

Note: (1)-(2) cluster year; (3)-(4) cluster ADM2. Baseline di gruppo/tercili omesse per collinearita'. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

*Metodo.*Water availability sono degli indicatori esterni di disponibilità idrica assegnati agli ADM2 e suddivisi per mediana o tercili. Le interazioni testano se la presenza di risorse idriche modula l'impatto.

Chapter 5

Conclusion

This thesis has explored how exposure to international maritime trade shapes the likelihood of conflict across Nigerian regions. The analysis combined information on port-level activity, maritime connectivity, and geography to build an index of local exposure to global trade shocks. The evidence points to a heterogeneous pattern: the effect of trade on conflict varies widely across space, sectors, and local resource conditions.

5.1 Main findings

Baseline and falsification. For commercial ports, the aggregate effect of the exposure index is small on average, but analysing the heterogeneity of the sample given the development of each ADM2 (proxied by night-lights), we discovered that in relatively less developed areas, higher exposure to trade is associated with lower conflict, while effects in more developed areas are near zero. When index is extentet to all ports (including those without commercial relevance), coefficients become weaker and statistically insignificant. This falsification supports the interpretation that the mechanism runs through international trade, in particolar access to international market, rather than mere proximity to coastal infrastructure.

Geographic heterogeneity. We studied different heterogeneity to study how the relationship varies across different dimension. Results show that there is a stabilizing effect stronger in the eastern belt than in the west. This spatial asymmetry matches economic patterns: eastern ADM2s are more rural, as shown by the proxy of nightlights, making them more sensitive to trade-induced welfare improvements.

Sector and trade flow heterogeneity. Result shows that the negative association is more pronounced for ports oriented toward the one that trade vegetables, as second most traded industry, and imports more with respect of what they export. This is consistent with our theory of consumption-smoothing channel.

Resource endowments (water). Interactions with water availability (a proxy for agricultural potential) indicate that the pacifying effect concentrates in resource-constrained areas. Where water availability is higher—hence greater agricultural

rents or competition—the relationship can weaken or even turn positive, suggesting that trade may intensify distributional tensions when valuable rents are at stake.

5.2 Interpretation

Overall, the findings support an opportunity-income channel, especially in poorer districts. Where communities gain easier access to markets or cheaper imported goods, the economic incentives to engage in violence tend to weaken. In contrast, where economic rents and pre-existing advantages are larger, trade may amplify inequalities and make consumption-related conflicts more likely. In short, trade is stabilising where markets fill and income opportunities, while it risks destabilising where it amplifies returns and contestable surpluses.

5.3 Limitations

While the findings are robust across specifications reported here, several caution should be taking in consideration: First measurement. CPT and LSCI are port indicators available for a shorter window, for which we mapped to the study period and the other port indicators, sectoral and flow classifications rely on IMF PortWatch assumption. Night-lights and water availability are proxies for income and agricultural potential rather than direct measures. Second, inverse distance to the nearest port abstracts from the situation of actual infrastructures; future work should embed explicit transport networks and travel times. Third conflict counts aggregate heterogeneous events; although the interpretation emphasizes consumption-related mechanisms, misclassification remains possible. Lastely, the design mitigates simultaneity by using lagged port shocks and fixed effects, and the "all-ports" falsification helps; still, unobserved time-varying shocks correlated with both exposure and conflict cannot be fully ruled out.

5.4 Policy implications

The heterogeneity of effects suggests that how a country integrates is as important as the volume of its trade. *Target lagging regions.* Investments that reduce effective distance to commercial ports for poorer inland districts can enhance the conflict-reducing benefits of trade.
Safeguard the welfare channel. Policies that keep essential import pipelines reliable and limit pass-through frictions help stabilize local prices and incomes.
textitManage rents where activity is high. In more developed or water-abundant areas, complement trade expansion with transparency, competition policy, and inclusive local fiscal instruments to curb rent capture and distributional conflict.
Shock-response tools. Temporary safety nets and price-stabilization measures during large external swings can preserve the pacifying effect of integration.

5.5 External validity and future research

Although Nigeria's coastal geography and regional fragmentation make it a compelling test case, external validity should be assessed cautiously. Extending the approach to other coastal African economies and to land-linked settings connected by corridors would clarify the role of transport networks. Three concrete extensions emerge: (i) network-based exposure, in fact replace straight-line distance with multimodal travel times and port-choice models; allow exposure to multiple ports and transshipment hubs. (ii) Micro-outcomes such as link exposure to prices, firm dynamics, and employment to pin down the welfare more directly. (iii) Institutional moderators: interact exposure with local governance quality, security force presence, and customary authority to test capacity-to-manage-distributional-tensions.

5.6 Contribution

Empirically, the thesis introduces a geography aware exposure metric that ties port level shocks and connectivity to local conflict dynamics, and documents sizable heterogeneity by development, location, sector, and resource endowment. Conceptually, it bridges the literature on external income shocks and the spatial diffusion of global markets: trade can pacify—but not everywhere and not unconditionally. Where integration expands livelihoods, conflict tends to recede; where it magnifies contestable rents, stability may not follow. Recognizing and acting on this conditionality is central to leveraging globalization for peace.

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Appendix

.1 conflict

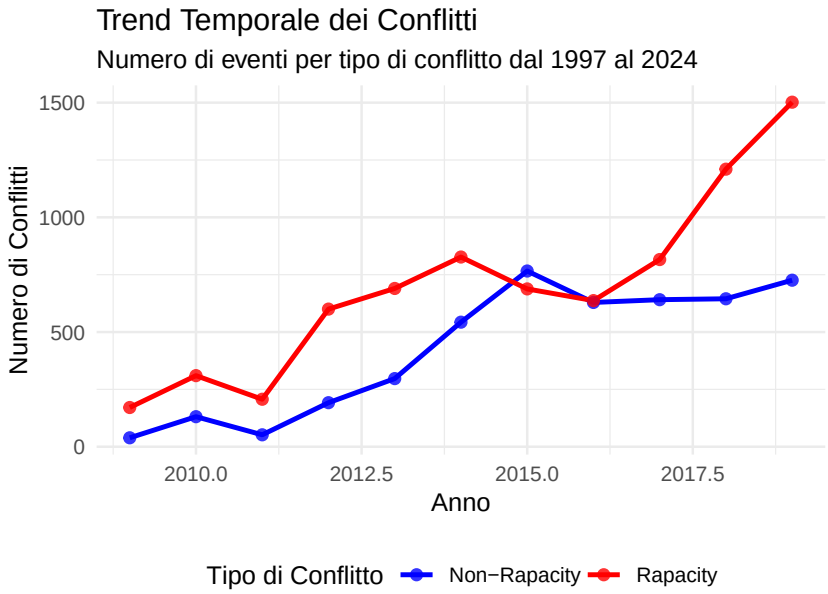


Figure .1 Trend of conflict (rapacity and non rapacity)

.2 Nightlights

.3 Spatial heterogeneity

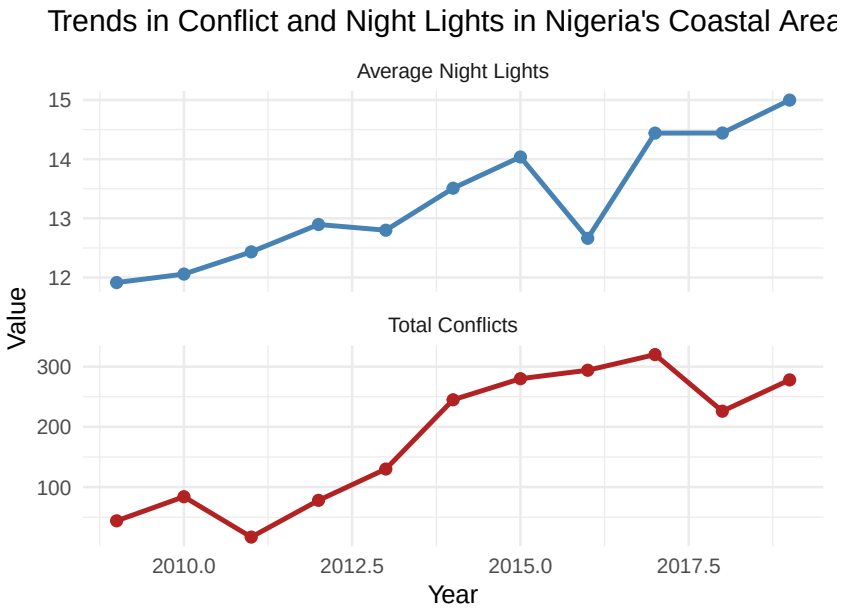


Figure .2 Conflict and nightlight

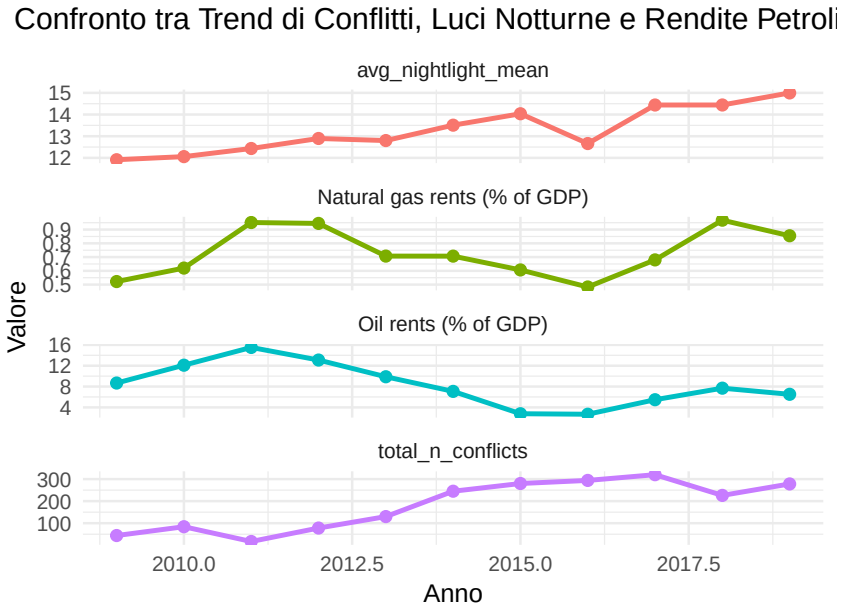


Figure .3 Confroning conflight, nightlight and different type of rent

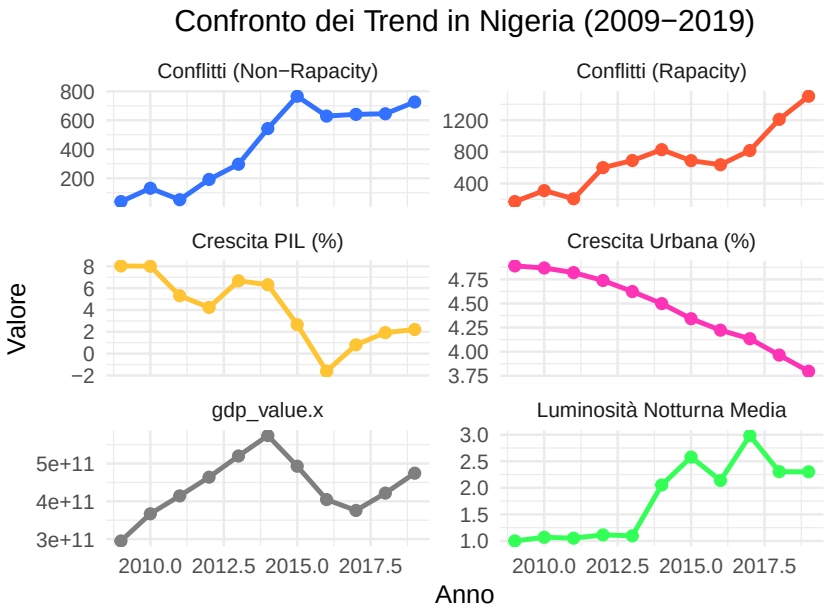


Figure .4 Confroning different type of conflight, nightlight and different type of rent

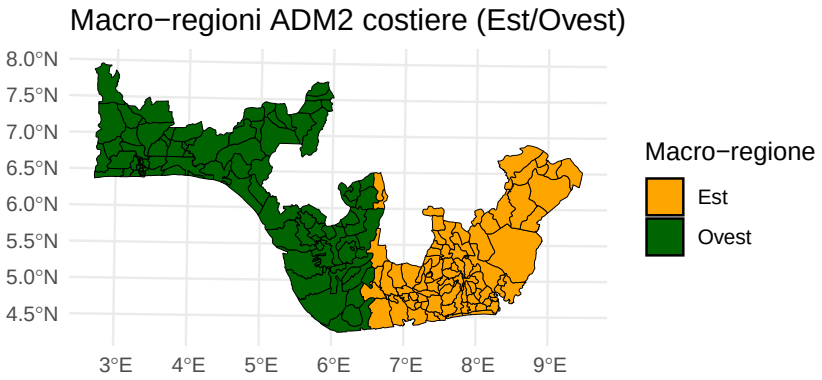


Figure .5 est and ovest

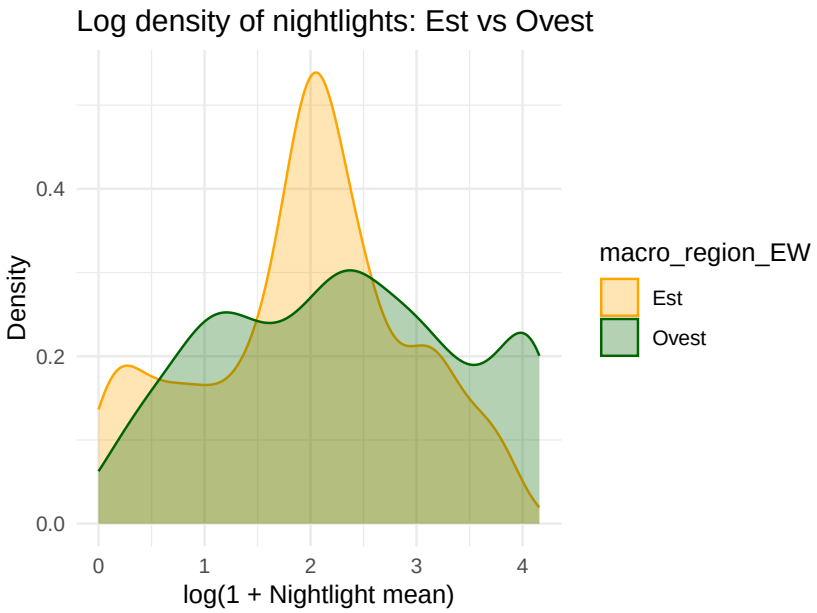


Figure .6 log density of nightlights

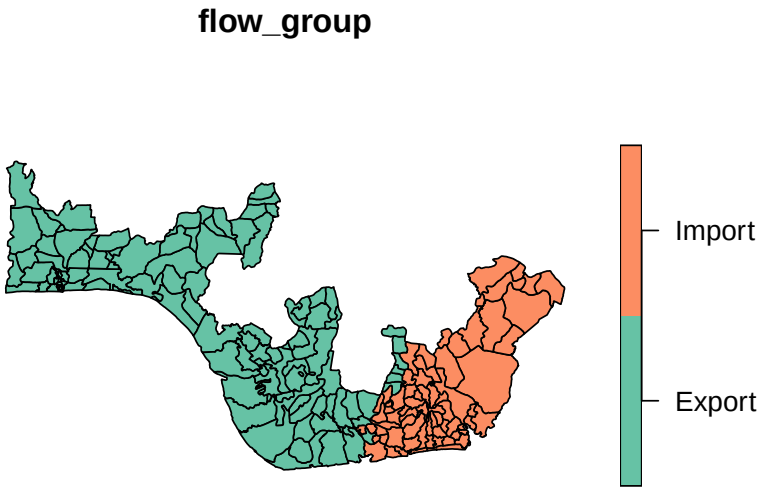


Figure .7 Import and export

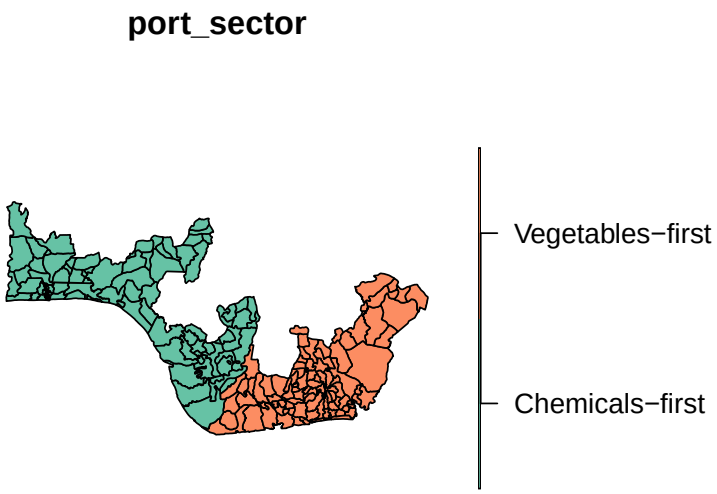


Figure .8 vegetable and chemicals